MIDTERM — ANSWER SHEET — SPRING 2022

EE3042 Energy Conversion Midterm Examination

Time Allowed: 75 minutes April 13th, 2022
Closed Notes and Books, Use only calculator, Not mobile phone
All necessary formulas are provided at the last page.

Student No. : ..... Name : .....SURNAME : .....Signature:.....

Part 1 (20p) Fill in the following blank(s) (2 points each) and tick the correct answers (5 to 10)

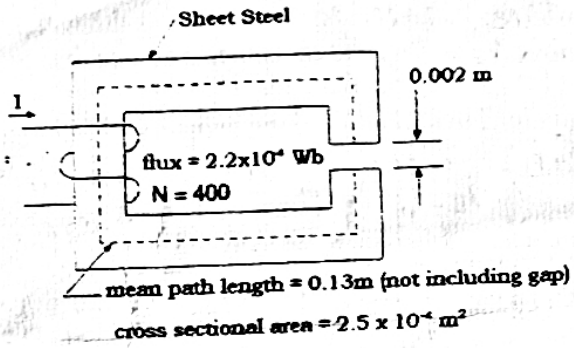
1. If a conductor is wound around a piece of metal, such as iron or steel, and current is passed through that conductor, a magnetic field is produced around this assembly. It is called electromagnet
2. In a rotating field (AC) generator, direct current from a separate source is passed through windings on the rotor by means of slip rings and brushes.
3. The armature windings are defined as the windings in which a voltage is induced, and the field windings are defined as the windings that produce the main magnetic flux in the machine.
4. The four equations that determine the behavior of a linear machine are 1. $e_{ind} = (\vec{v} \times \vec{B}) \cdot \vec{l}$ 2. $\vec{F} = i(\vec{l} \times \vec{B})$, 3. KVL ($v_B - iR - e_{ind} = 0$), and 4. $F_{net} = ma$ (Newton's law)
5. Hysteresis loss minimally depends on
   - (a) volume of the magnetic material
   - ✓(b) ambient temperature
   - (c) coefficient of material
   - (d) applied frequency
6. The best approach to reduce eddy currents is to
   - ✓(a) increase resistance of core (by using laminated core) (thin sheets of metal insulated from each other)
   - (b) handle thick wire
   - (c) reduce hysteresis loss
   - (d) increase core permeability
7. The leakage flux in a transformer depends upon the value of (Core flux remains constant)
   - (a) frequency
   - (b) mutual flux
   - ✓(c) load current
   - (d) applied voltage

   (Primary and secondary fluxes increase. Their difference (core flux) is unchanged. Hence Leakage flux with the load current)
8. Which of the following losses varies with the load in the transformer?
   - ✓(a) copper loss
   - (b) core loss
   - (c) both core and copper loss
   - (d) none of the above
9. Transformer core are laminated in order to
   - (a) Reduce hysteresis loss
   - (b) Reduce hysteresis & eddy current loss
   - ✓(c) Minimize eddy current loss
   - (d) Copper loss
10. For a constant primary voltage, how does the secondary terminal voltage of a transformer change when the power factor of the capacitive load increases from 0.7 to 0.9 with the load current remaining unchanged?
    - (a) voltage increases
    - (b) voltage decreases
    - (c) voltage remains unchanged
    - (d) voltage will rise, pass maximum and then decrease

    When decreasing/increasing reactive power (Q) the real power remains unchanged. $P = V_s I_s \cos\theta$ [1] where I_s → constant. Since $\cos\theta$ increases, $Q = V_s I_s \sin\theta$ will decrease. Because θ decreases. from [1] $\frac{P}{I_s} = \text{constant} = V_s \cos\theta$. Hence V_s decreases. 1/1.5

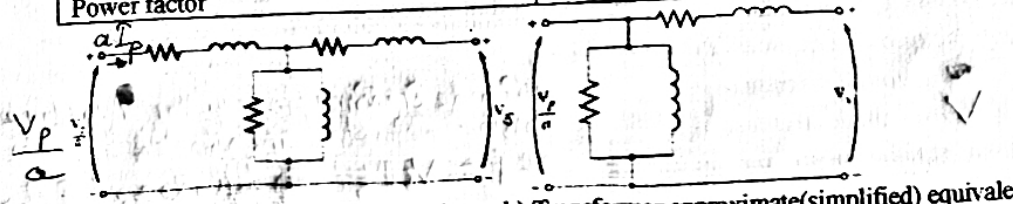
Part 2 (80p) Solve the following problems

1. (10p) A 200-turn coil is rotating in a magnetic field of magnitude B = 0.250T at a frequency of 60Hz. The area of the coil A = 5.00×10^{-2} m^2. What is the maximum emf in the coil?
2. (15p) Assume that the voltage applied to a load is $V = 220\angle 45°$ V and the current flowing through the load is $I = 3\angle 25°$ A.
   (a) Calculate the complex power S consumed by this load.
   (b) Is this load inductive or capacitive?
   (c) Calculate the power factor of this load.
3. (10p) Suppose that the voltage applied to a load is V = 200∠-30° V and the current flowing through the load is I = 5∠15° A. The complex power S consumed by the load, type of the load, and the power factor of the load are
4. (15p) Find the current and flux density for the air gap in the magnetic circuit (Figure 1) in which the relative permeability is 8000.



5. (40p) (2.1) The A 100-kVA 8000/277-V distribution transformer has the following resistances and reactances:
   $R_p = 4.5\ \Omega$, $R_s = 0.0045\Omega$, $X_p = 6.5\Omega$, $X_s = 0.0055\Omega$, $R_c = 52k\Omega$, $X_M = 10.2k\Omega$
   The excitation branch impedances are given referred to the high-voltage side of the transformer.
   (10) (a) Find the equivalent circuit of this transformer referred to the low-voltage side.
   (10p) (b) Find the per-unit equivalent circuit of this transformer.
   (10p) (c) Assume that this transformer is supplying rated load at 277 V and 0.85 PF lagging. Using simplified transfermer equivalent circuit, determine transformer's input voltage? What is its voltage regulation in percent?
   (5p) (d) What are the copper losses and core losses in this transformer under the conditions of part (c)?
   (5p) (e) What is the transformer's efficiency under the conditions of part (c)?

Complex Power	$S = VI^*$
Real Power	$P = VI\cos(\theta)$
Reactive Power	$Q = VI\sin(\theta)$
Power Converted to mechanical	$P = E_A I_A$
Motor shaft power	$Pout = w\,\tau$
Generated voltage (bemf)	$E_A = K\emptyset w$
Core Reluctance	$\mathcal{R} = \frac{l}{\mu_o \mu_r A}$
Permeability in free space (μ_o)	$4\pi * 10^{-7}$
Voltage Regulation	$VR = \frac{V_{no\,load} - V_{FL}}{V_{FL}}$
Power loss	$P = I^2R = \frac{V^2}{R}$
Power factor	$pf = \cos\theta$



a) Transformer equivalent circuit b) Transformer approximate(simplified) equivalent circuit

MIDTERM ANSWER SHEET SPRING 2022

Part II

(10p) 1) 200-turn coil
$B = 0{,}250\ T$
$f = 60\ Hz$
$A = 5\times10^{-2}\ m^2$ (of coil)
$E_{emf} = ?$

$$Emf = \frac{d\phi}{dt} = NA\frac{d\vec{B}}{dt}$$ where

$\vec{B} = B\cos\omega t$

$$\Rightarrow Emf = NAB\frac{d\cos\omega t}{dt} = -NAB\,\omega\sin\omega t$$

$$\Rightarrow Emf = -NAB(-2\pi f)\sin(2\pi f t)$$

where sine function changes between +1 and −1. Hence $Emf_{max} = NAB(2\pi f)$

$$\Rightarrow Emf_{max} = (200)(5\times10^{-2})(0{,}250)\left(2\pi\times60\frac{1}{s}\right)$$ (1 second)

$$\cong 942.5\ V$$

or (2nd solution)

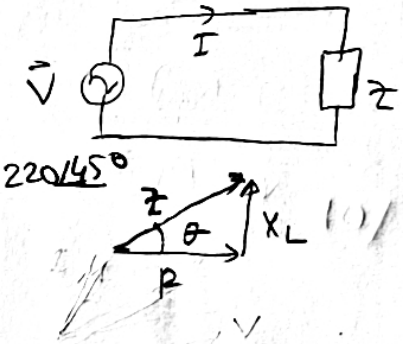
$\phi = BA = (0{,}25)(5\times10^{-2}) = 0{,}0125\ Wb$

$Emf = N\phi\omega = N\phi(2\pi f)$

$= (200)(0{,}0125)(2\pi\times60)$

$= 942{,}5\ V$

2) (15p) $V = 220\angle45°$, $I = 3\angle25°$
5/a) S ?
5/b) Load type?
5/c) pf = ?

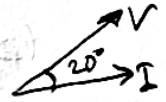


$\vec{V} = 220\angle45°$

a) $\vec{S} = \vec{V}\vec{I}^{*} = (220\angle45°)(3\angle-25°) = 660\angle20°$

b) $Z = \frac{V}{I} = \frac{220\angle45}{3\angle25} = 73.33\angle20°$

Inductive load

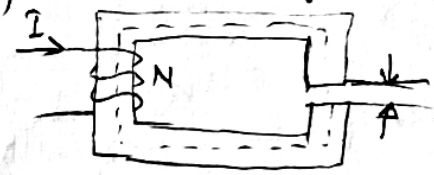


where $\theta = 20°$

c) $pf = \cos\theta = \cos(20°) = 0{,}94$

3) ~~[illegible]~~ canceled

4) (15p) i and B_g? $\mu_r = 8000$ steel



$N = 400$ turns, $A = 2.5\times10^{-4}\ m^2$
$\phi = 2{,}2\times10^{-4}\ Wb$
$l_c = 0{,}13\ m$. (not including air gap length)

$$B_g = \frac{\phi_L}{A} = \frac{2{,}2\times10^{-4}}{(2{,}5\times10^{-4})} = 0{,}88\ T$$

$$\mathcal{F} = NI = \phi\mathcal{R} \Rightarrow I = \frac{\phi\mathcal{R}}{N} = \frac{\phi(\mathcal{R}_g+\mathcal{R}_c)}{N}$$

$$\mathcal{R} = \mathcal{R}_g + \mathcal{R}_c = \frac{l_g}{\mu_0 A} + \frac{l_c}{\mu_0\mu_r A} = \frac{0{,}002}{4\pi\times10^{-7}(2.5\times10^{-4})} + \frac{0{,}13}{(4\pi\times10^{-7})(8{,}000)\times(2{,}5\times10^{-4})}$$

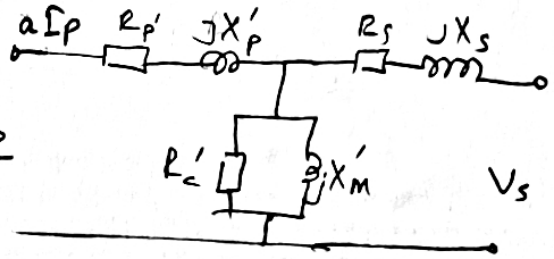
$\mathcal{R} \cong 6{,}42\times10^{6}$ A turns/Wb

$$I = \frac{\phi\mathcal{R}}{N} = \frac{(2{,}2\times10^{-4})(6{,}42\times10^{6})}{400} = 3{,}53\ A$$ 3/5

MIDTERM ANSWER SHEET SPRING 2022

5.) 100kVA, 8000/277-V distribution tr.

(40p)



$R_p = 4.5\Omega$, $R_s = 0,0045\Omega$.

$X_p = 6.5\Omega$, $X_s = 0,0055\Omega$, $R_c = 52k\Omega$

$X_M = 10,2k\Omega$

10/ a) The equivalent circuit referred to the LVS.

$a = \frac{8000}{277} = 28,88$

$R_p' = \frac{R_p}{a^2} = \frac{4,5}{(28,88)^2} = 0,0054\Omega$.

$X_p' = \frac{X_p}{a^2} = \frac{6,5}{(28,88)^2} = 0,0078\Omega$

$R_c' = \frac{R_c}{a^2} = \frac{52k\Omega}{(28,8)^2} = 62\Omega$.

$X_M' = \frac{X_M}{a^2} = \frac{10,2k\Omega}{(28,88)^2} = 12,23\Omega$

$R_s = 0,0045\Omega$, $X_s = 0,0055\Omega$

b) 10p (rated)

$S_{base} = 100kVA$, V_{base} (rated) $= 277$, I_{base} (rated) $= \frac{S_{base}}{V_{base}} = \frac{100kVA}{277} = 361A$

$\Rightarrow Z_{base} = \frac{V_{base}}{I_{base}} = \frac{277}{361} = 0,767\Omega$. OR $Z_{base} = \frac{V_{base}^2}{|S|} = 0,767$

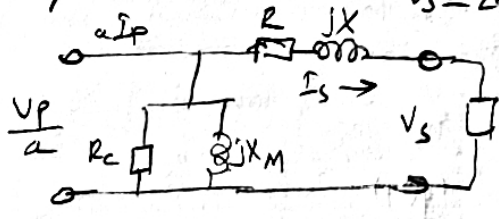
PU $\Rightarrow$ Z_{pu} referred to secondary $= \frac{Z_{actual}}{Z_{base}} \Rightarrow Z_{pu}$ primary $= \frac{R_p' + jX_p'}{Z_{base}} = \frac{0,0054 + j0,0078}{0,767} = 0,007 + j0,01$ (R_p', X_p')

PU $\Rightarrow Z_{pu}$ secondary $= \frac{0,0045 + j0,0055}{0,767} = 0,00587 + j0,00717$ (R_s, X_s)

Another method: PU values can be obtained by taking primary and secondary base quantities separately for each side.

$R_{c_{pu}} = \frac{R_c'}{Z_{base}} = \frac{62}{0,767} = 80,83\Omega$, $X_{M_{pu}} = \frac{X_M'}{0,767} = \frac{12.23}{0,767} = 15,945\Omega$.

c) 10/ Rated load at $V_T = 277V$, $pf = 0,85$ lagging



$Vs = 277V$

$I_s = \frac{V_s}{Z_L} = \frac{277}{0,767\angle 31,8} = 361\angle -31,8°$ (or $I_s = \frac{S\angle\theta}{V_s}$)

$V_p' = \frac{V_p}{a} = V_s + (R_{EQ} + jX_{EQ})I_s = 277\angle 0° + [(0,0054 + 0,0045) + j(0,0078 + 0,0055)]361\angle -31,8$

($0,0099$, $0,0133$)

$V_p' = 277\angle 0° + (0,0166\angle 53.34°)(361\angle -31,8)$

$= 277\angle 0° + 5.993\angle 21.54° = 282.58\angle 0,45°$

$VR = \frac{V_p' - V_s}{V_s} \times 100 = 2\%$

$VR = \frac{282,58 - 277}{277} \times 100$ ↑ 4/5

5-continued) 5/ d) $P_{cu} = (R_{EQ})(I_s)^2 = (0,0099)(361)^2 = 1290.2\,W$

$P_{core} = \frac{V_p'^2}{R_c} = \frac{(282.58)^2}{62\,\Omega} = 1.288\,kW$

5/ e) $P_{out} = V_s I_s \cos\theta = (277)(361)\cos(-31.8) \cong 85\,kW$

or $P_{out} = S\cos\theta = (100\,kVA)(0,85) = 85\,kW$

$\eta = \frac{P_{out}}{P_{out} + P_{cu} + P_{core}} \times 100 = \frac{85\,kW}{85\,kW + 1290.2 + 1288.6} \cong 97.3\%$